# 8 – Cont. Bivariate Dist

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# Recap: Multivariate Integral & its pdf

The **joint cumulative density function (joint CDF)** of $X$ and $Y$ is given by

$$F(x,y) = \mathbb{P}(X \le x, Y \le y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(s,t)\, dt\, ds,$$

where $f(x,y)$ is the joint PDF of $X$ and $Y$.

The respective **marginal PDF** of continuous-type random variables $X$ and $Y$ are given by

- $$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy,$$
- $$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx.$$

# Recap: Multivariate Integral & its pdf

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where $f(x,y)$ is the joint PDF of $X$ and $Y$.

Suppose X and Y are **independent**.

If X has a pdf of g(x) and Y has a pdf of h(y),
then **f(x,y) = g(x)h(y)**

**Problem 1 (Joint PDF, Marginal PDF)** Let $f(x,y) = (3/16)xy^2$, $0 \le x \le 2$, $0 \le y \le 2$, be the joint PDF of $X$ and $Y$.

(a) Find $f_X(x)$ and $f_Y(y)$, the marginal PDF of $X$ and $Y$ respectively.

(b) Are the two random variables independent? As in the discrete case, two continuous-type random variables $X$ and $Y$ are independent provided $f(x,y) = f_X(x)f_Y(y)$.

(c) Compute the mean $\mu_X$ and variance $\sigma_X^2$ of $X$.

(d) Find $\mathbb{P}(X \le Y)$.

(c)

$$\mu_X = \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^2 x \cdot \frac{x}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} = \frac{4}{3}.$$

$$\sigma_X^2 = \mathbb{E}[X^2] - \mu_X^2 = \int_{\infty}^{\infty} x^2 f_X(x)\,dx - \left(\frac{4}{3}\right)^2 = \int_0^2 x^2 \cdot \frac{x}{2}\,dx - \frac{16}{9} = \left[\frac{x^4}{8}\right]_0^2 - \frac{16}{9} = \frac{2}{9}.$$

$$\begin{aligned}
\mathbb{P}(X \le Y) &= \int_0^2 \int_x^2 \frac{3}{16} x y^2\,dy\,dx \\
&= \frac{3}{16}\int_0^2 x \left[\frac{y^3}{3}\right]_x^2 dx \\
&= \frac{3}{16}\int_0^2 x \left(\frac{8}{3} - \frac{x^3}{3}\right) dx \\
&= \frac{1}{16}\int_0^2 8x - x^4\,dx \\
&= \frac{1}{16}\left[\frac{8x^2}{2} - \frac{x^5}{5}\right]_0^2 \\
&= \frac{3}{5}.
\end{aligned}$$

**Problem 2 (Joint PDF, Marginal PDF, Conditional PDF)** Let $f(x, y) = 1/40$, $0 \leq x \leq 10$, $10 - x \leq y \leq 14 - x$ be the joint PDF of $X$ and $Y$.

(a) Sketch the region of the points $(x, y)$ satisfying the inequalities $0 \leq x \leq 10$, and $10 - x \leq y \leq 14 - x$.

(b) Find $f_X(x)$, the marginal PDF of $X$.

(c) Determine $h(y|x)$, the conditional PDF of $Y$, given that $X = x$.

(d) Calculate $\mathbb{E}[Y|X = x]$, the conditional mean of $Y$, given that $X = x$.

(d) The conditional mean of $Y$ given $X = x$ is

$$\begin{aligned}
\mu_{Y|x} &= \mathbb{E}[Y|X = x] \\
&= \int_{10-x}^{14-x} y \cdot \frac{1}{4}\, dy \\
&= \frac{1}{4}\left[\frac{y^2}{2}\right]_{10-x}^{14-x} \\
&= \frac{1}{4}\left[\frac{(14-x)^2}{2} - \frac{(10-x)^2}{2}\right] \\
&= 12 - x,
\end{aligned}$$

## Problem 3 (Conditional PDF, Conditional Expectation)

Let $X$ and $Y$ be continuous random variables with joint PDF

$$f(x,y) = \begin{cases} x + \frac{3}{2}y^2, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the conditional PDF $f(x|y)$ for all $x, y$.

(b) Compute the conditional expectation $E[X|Y=y]$ for all $y$.

(c) Find the conditional probabilities (i) $P\left(X \le \frac{1}{2}|Y = \frac{1}{2}\right)$ and (ii) $P\left(\frac{1}{4} \le X \le \frac{3}{4}|Y = \frac{1}{2}\right)$.

(b) We compute

$$E[X|Y=y] = \int_0^1 x f(x|y)dx = \int_0^1 x\left(\frac{2x+3y^2}{3y^2+1}\right)dx = \frac{9y^2+4}{6(3y^2+1)}$$

(c) We compute

$$P\left(X \le \frac{1}{2}\middle|Y = \frac{1}{2}\right) = \int_{-\infty}^{1/2} f\left(t\middle|\frac{1}{2}\right)dt = \int_0^{1/2} \frac{2x+\frac{3}{4}}{\frac{3}{4}+1}dx = \frac{5}{14},$$

$$P\left(\frac{1}{4} \le X \le \frac{3}{4}\middle|Y = \frac{1}{2}\right) = \int_{1/4}^{3/4} \frac{2x+\frac{3}{4}}{\frac{3}{4}+1}dx = \frac{1}{2}.$$

**Problem 4 (Joint PDF, Marginal PDF, Conditional probability)**
Let $X$ and $Y$ have the joint PDF $f(x, y) = cx(1 - y)$, $0 < y < 1$, and $0 < x < 1 - y$, where $c$ is a constant.

(a) Determine $c$.

(b) Compute $\mathbb{P}(Y < X \mid X \leq 1/4)$.

$$\boxed{\mathbb{P}(Y < X|X \leq 1/4) = \frac{\mathbb{P}(0 < Y < X \leq 1/4)}{\mathbb{P}(X \leq 1/4)} = \frac{\frac{29}{24\times 32}}{\frac{31}{32\times 8}} = \frac{29}{31 \times 3} = \frac{29}{93}.}$$

# Bonus Qn: Finals AY22/23 3(b)

(b) In a number game, two participants make guesses of $X$ and $Y$ respectively. The joint PDF of $X$ and $Y$ is **uniform** (i.e. constant) on the region $1 \leq x \leq 10$, $1 \leq y \leq 9$. If $|X - Y| < 1$, then the two participants will be asked to guess again. What is the probability that they will be asked to guess again?

# Bonus Qn: 2 Indept. Exponentials

Suppose we have 2 independent variables X and Y and they are exponentially distributed. X has a scale parameter of 1/2 and Y has a scale parameter of 1/3 with the following pdf:

$$f(x) = 2e^{-2x} \qquad\qquad g(y) = 3e^{-3y}$$

What is Pr{X – Y >1}?